

B.Tech Degree IV Semester Examination April 2012

ME 406 MANUFACTURING PROCESS

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) List any five applications of casting.
(b) Explain the significance of gating ratio.
(c) Write a brief note on casting cleaning.
(d) Explain die casting.
(e) Distinguish between hot working and cold working.
(f) Discuss the principle of rolling.
(g) Explain thermit welding.
(h) Explain the principle of shielded arc welding.

PART B

(4 x 15 = 60)

- II. Explain in detail about the desirable properties of a moulding sand.

OR

- III. (a) Give a brief account on the forces acting on moulding flasks.
(b) Explain the different types of cores.

- IV. Describe in detail about the various casting defects and also how it can be eliminated.

OR

- V. Write notes on:

- (i) Centrifugal casting
(ii) Shell moulding.

- VI. Explain the following with appropriate sketches:

- (i) Press forging
(ii) Forging design

OR

- VII. Explain the following operations:

- (i) Tube extrusion
(ii) Spinning
(iii) Drawing

- VIII. Explain gas welding in detail, indicating clearly the different gas combinations used, types of flames and applications.

OR

- IX. (a) Distinguish between brazing and soldering.
(b) Explain TIG welding.